

Lab Assignment-2

Course: Probability and Statistics

Course code: BMAT202P

Date: 04-03-2026

Answer all the questions using R markdown and upload in VTOP.

1. The following table gives the weight (x) (in 1000 lbs.) and highway fuel efficiency (y) (in miles/gallon) for a sample of 13 cars.

<i>Vehicle</i>	<i>x</i>	<i>y</i>
chevrolet camaro	3.545	30
dodge neon	2.6	32
honda accord	3.245	30
lincoln continental	3.93	24
oldsmobile aurora	3.995	26
pontiac grand am	3.115	30
mitsubishi eclipse	3.235	33
bmw 3-series	3.225	27
honda civic	2.44	37
toyota camry	3.24	32
hyundai accent	2.29	37
mazda protégé	2.5	34
cadillac deville	4.02	26

Find the correlation between x and y.

2. Find the Correlation between below data.

ENJOY	BUY	READ
4	16	6
15	19	13
1	0	1
11	19	13
13	25	12
19	24	11
6	22	7
10	21	8
15	13	12
3	7	4
11	28	15
20	31	14
7	4	7
11	26	14
10	11	9
6	12	5
7	14	7
18	16	12

8	20	10
2	13	6
7	12	9
12	13	23
13	22	9
15	19	13
4	12	9
3	10	5
9	7	7
7	22	8
10	7	8
2	0	2
15	16	7
1	17	6
3	11	9
6	5	9
13	29	15
15	29	11
16	20	9
14	16	7
1	3	2
8	8	10

3. For the given details viz. Sector wise Number of Factories, Productive Capital, No. of Employees, Total Output and Net Value Added Fit the Multiple Regression and interpret your result. Assume the variables as Dependent and Independent according to your requirement/description.

Area	Y	X1	X2
1	110	30	11
2	80	40	10
3	70	20	7
4	102	50	15
5	105	60	19
6	90	40	12
7	70	20	8
8	120	60	14

4. Use the Life Satisfaction dataset to fit the regression equation.

import kagglehub
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path = kagglehub.dataset_download("unsdsn/world-happiness")
print("Path to dataset files:", path)

5. If 10% of the Screws produced by an automatic machine are defective, find the probability that out of 20 screws selected at random, there are (i) Exactly 2 defective (ii) At least 2 defectives (iii) Between 1 and 3 defectives (inclusive) (i) Exactly 2 defective.
6. For a random variable X with a binomial (20,1/2) distribution, find the following probabilities (i). Find $P(X < 4)$ (ii). Find $P(X > 11)$ (iii). Find $P(1 \leq X \leq 8)$.

6. Let X be the number of heads in 10 tosses of a fair coin. (i) Find the probability of getting at least 5 heads (that is, 5 or more). (ii) Find the probability of getting exactly 5 heads. (iii) Find the probability of getting between 4 and 6 heads, inclusive.
7. A recent national study showed that approximately 55.8% of college students have used Google as a source in at least one of their term papers. Let X equal the number of students in have used Google as a source:
 - i. How is X distributed?
 - ii. Sketch the probability mass function (roughly)
 - iii. Sketch the cumulative distribution function (roughly).
 - iv. Find the probability that X is equal to 17
 - v. Find the probability that X is at most 13.
 - vi. Find the probability that X is bigger than 11.
 - vii. Find the probability that X is at least 15.
 - viii. Find the probability that X is between 16 and 19,
 - ix. Give the mean of X , denoted $E(X)$.
 - x. Give the variance of X .
 - xi. Give the standard deviation of X .
 - xii. Find $E(4X + 51.324)$